

Feature Importance Report — Lung Full FOV

Classification Performance

Metric	Value
CV Accuracy (5-fold)	0.518 ± 0.015
CV Weighted F1	0.504 ± 0.015
Baseline (majority class)	0.090
Beats baseline	True
Number of classes	20
Samples	11898
Features	47

Modality Split

Modality	Gini Importance	Permutation Importance
Image features (24)	70.1%	100.0%
Expression features (23)	29.9%	0.0%

Top-20 Features (by Permutation Importance)

Rank	Feature	Gini Importance	Permutation (mean ± std)
1	compartment_boundary_distance	0.2147	0.4557 ± 0.0020
2	microenv_nndist_5	0.0320	0.0004 ± 0.0002
3	microenv_nndist_4	0.0289	0.0000 ± 0.0000
4	microenv_nndist_3	0.0279	0.0001 ± 0.0000
5	microenv_nndist_2	0.0276	0.0001 ± 0.0000
6	microenv_nndist_1	0.0274	0.0000 ± 0.0000
7	texture_energy	0.0265	0.0000 ± 0.0000
8	texture_homogeneity	0.0262	0.0000 ± 0.0000
9	morph_perimeter	0.0246	0.0000 ± 0.0000
10	morph_eccentricity	0.0242	0.0000 ± 0.0000
11	morph_equiv_diameter	0.0240	0.0000 ± 0.0000
12	texture_dissimilarity	0.0239	0.0000 ± 0.0000
13	morph_area	0.0238	0.0000 ± 0.0000
14	morph_extent	0.0234	0.0000 ± 0.0000
15	morph_solidity	0.0232	0.0000 ± 0.0000
16	texture_correlation	0.0230	0.0000 ± 0.0000
17	texture_contrast	0.0229	0.0000 ± 0.0000
18	expr_total_counts	0.0218	0.0000 ± 0.0000

19	expr_mean	0.0217	0.0000 ± 0.0000
20	compartment_intensity_gradient	0.0217	0.0000 ± 0.0000

Cluster Distribution

Cluster	Count	Percentage
0	1069	9.0%
1	1001	8.4%
2	971	8.2%
3	924	7.8%
4	684	5.7%
5	672	5.6%
6	624	5.2%
7	615	5.2%
8	610	5.1%
9	585	4.9%
10	567	4.8%
11	546	4.6%
12	470	4.0%
13	461	3.9%
14	456	3.8%
15	419	3.5%
16	402	3.4%
17	388	3.3%
18	265	2.2%
19	169	1.4%

Generated Plots

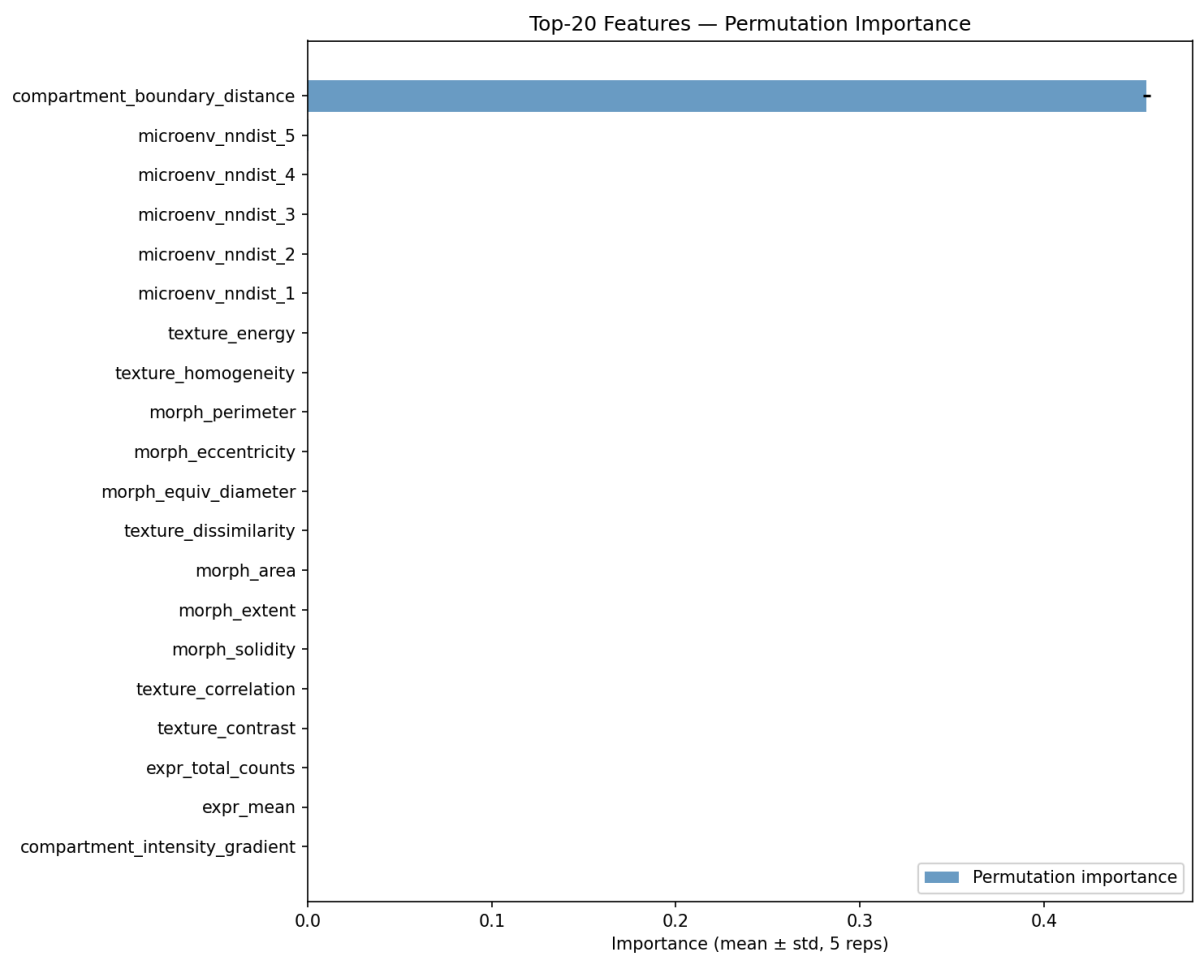


Figure 1: Top-20 Features

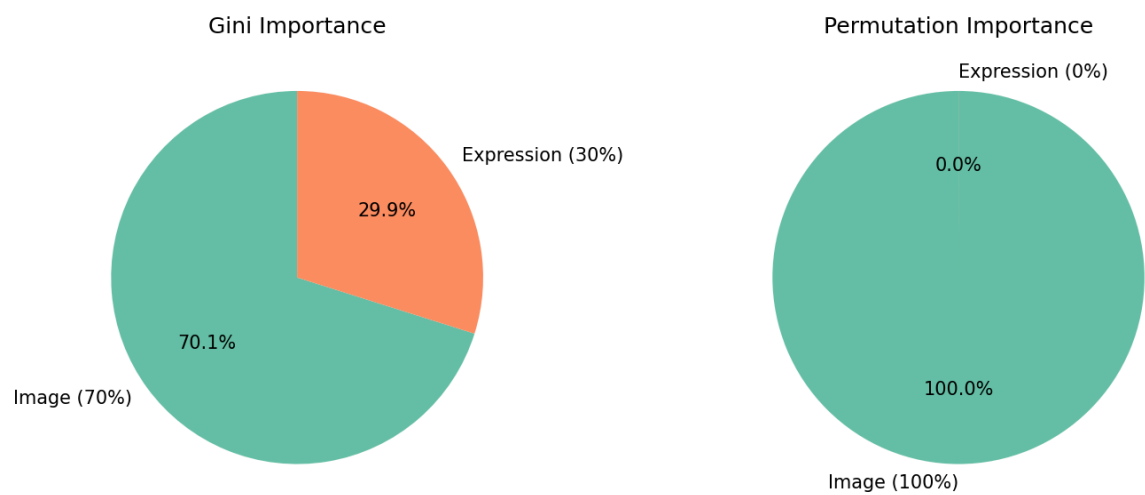


Figure 2: Importance by Modality

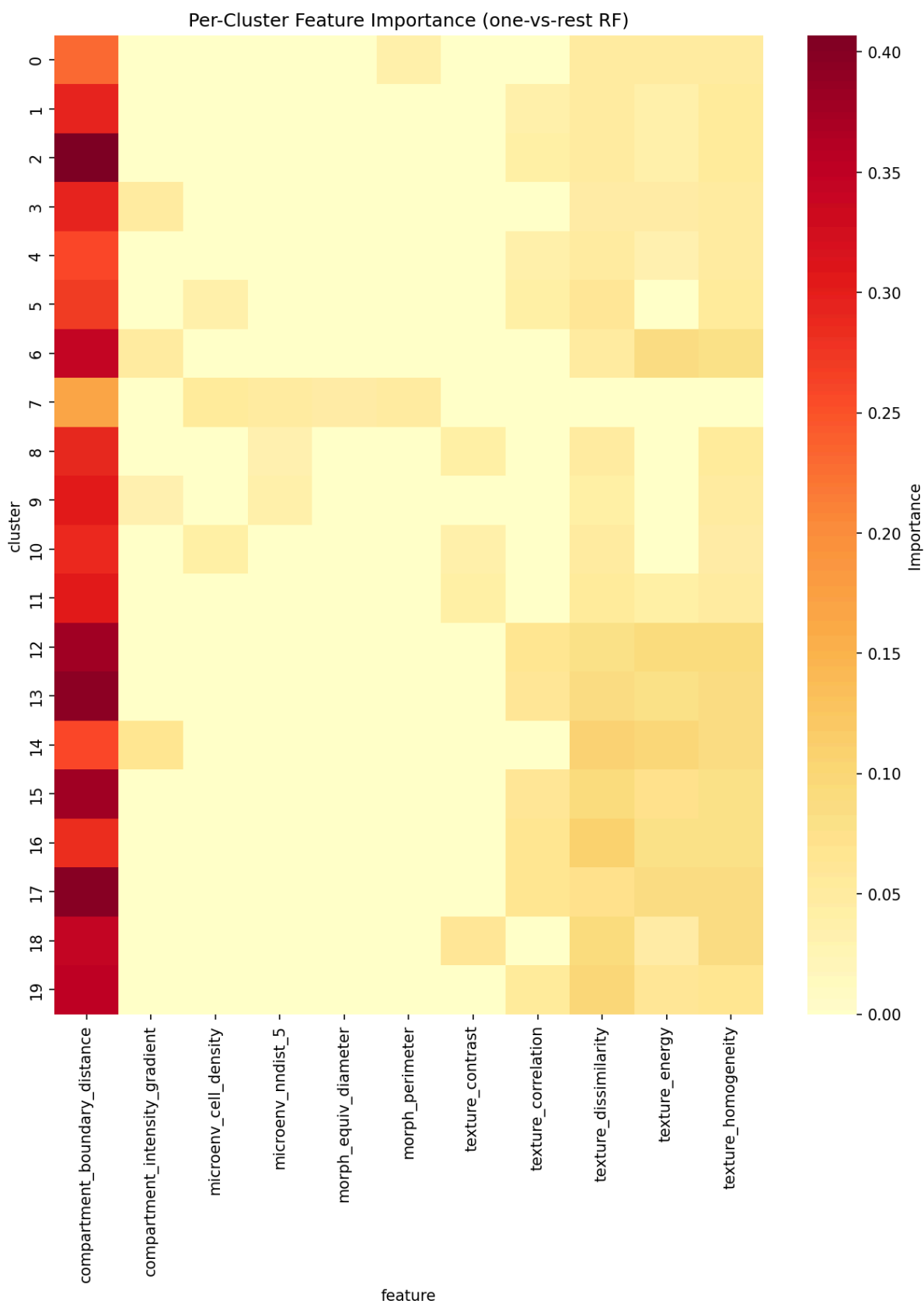


Figure 3: Per-Cluster Importance

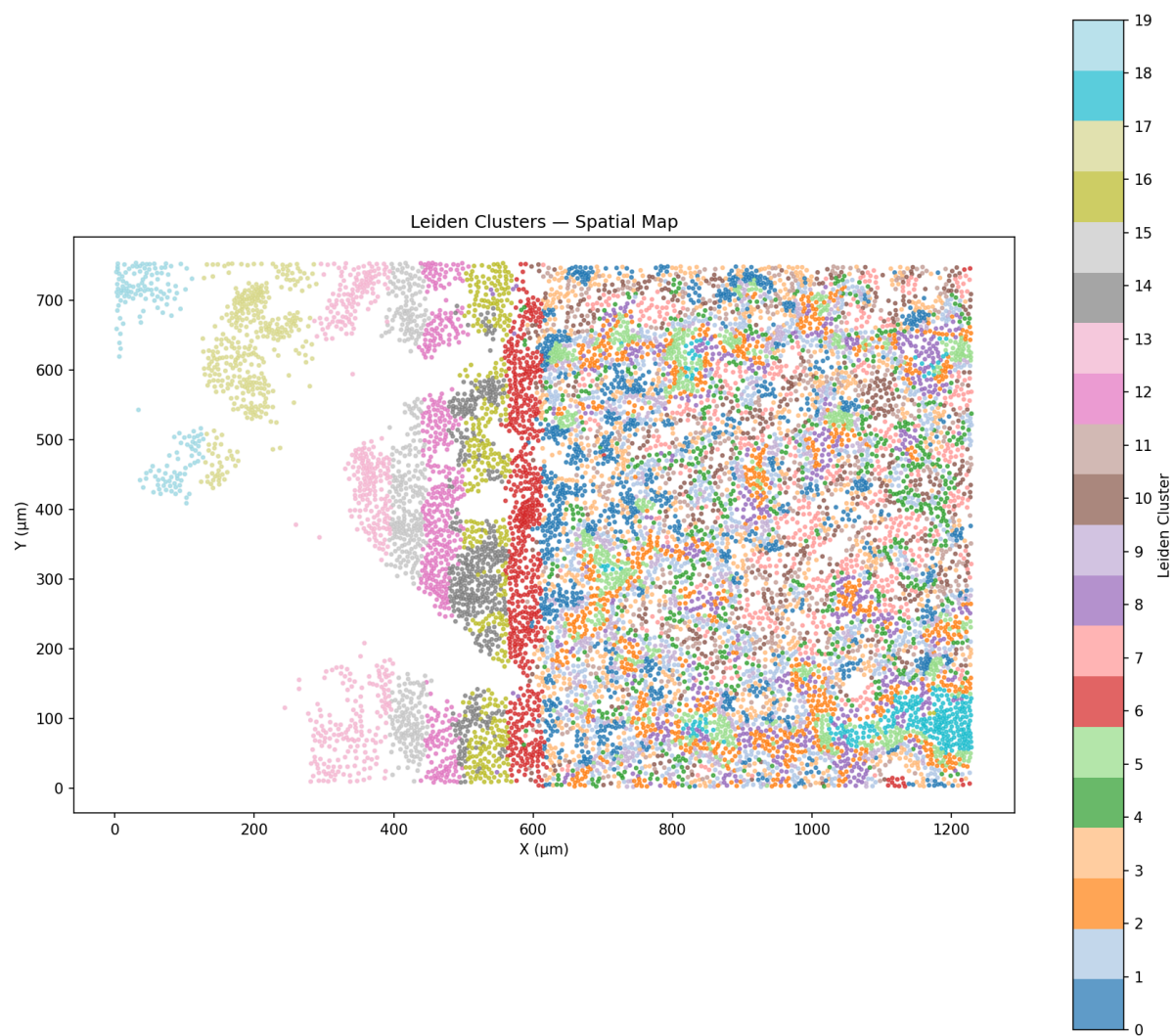


Figure 4: Spatial Clusters

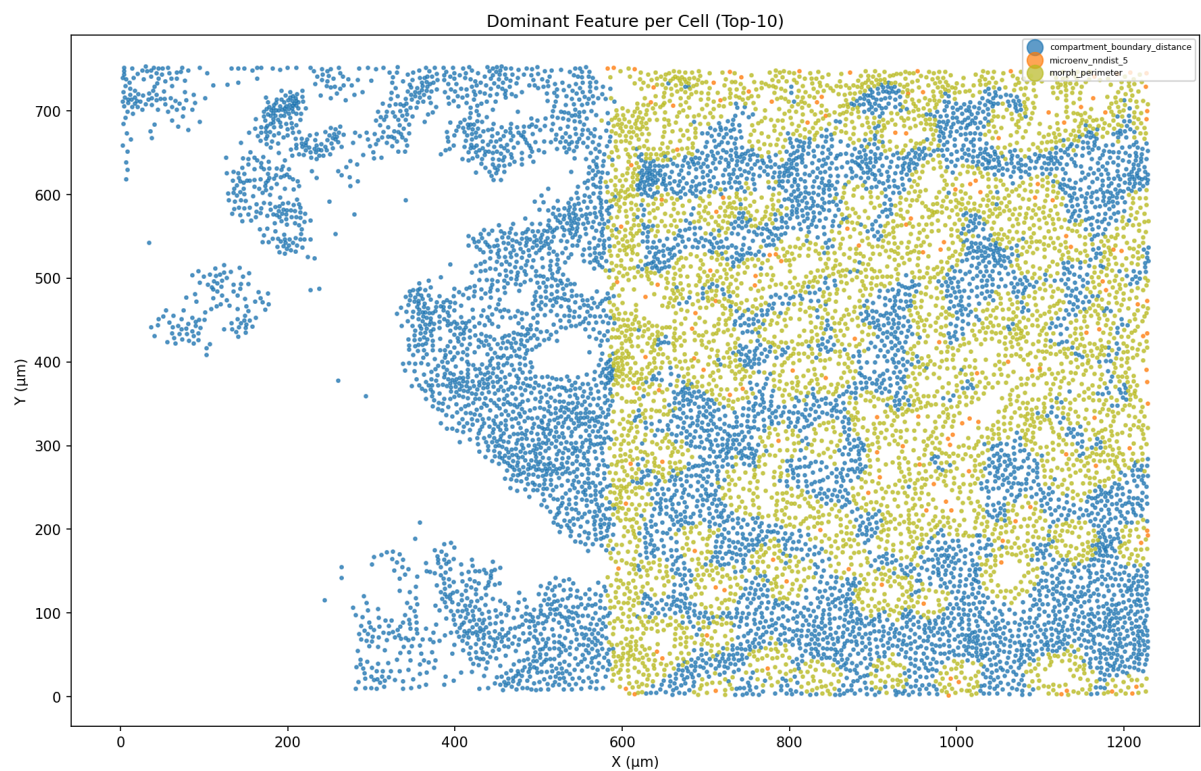


Figure 5: Dominant Feature per Cell

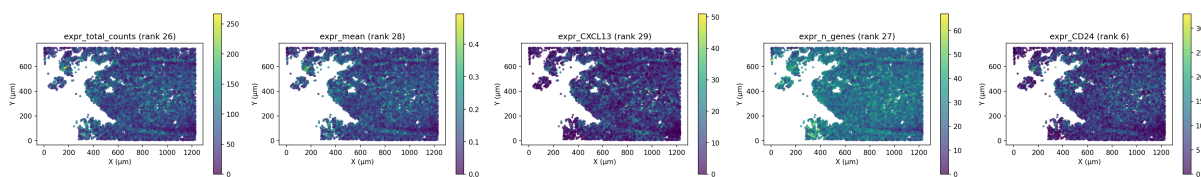


Figure 6: Top-5 Spatial Gene Expression

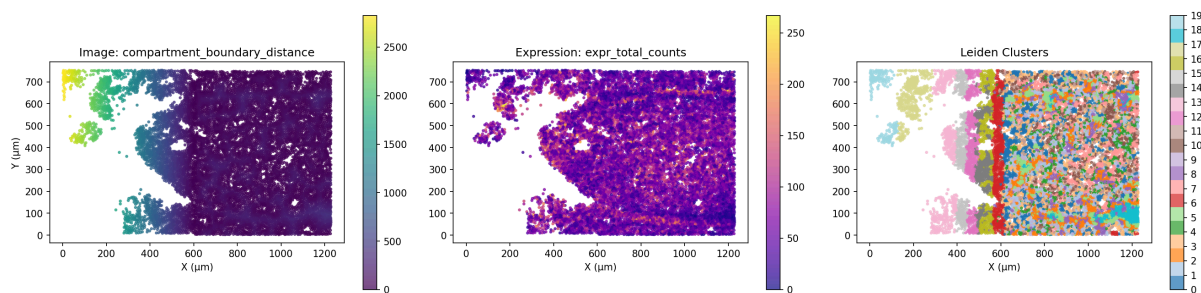


Figure 7: 3-Panel Comparison

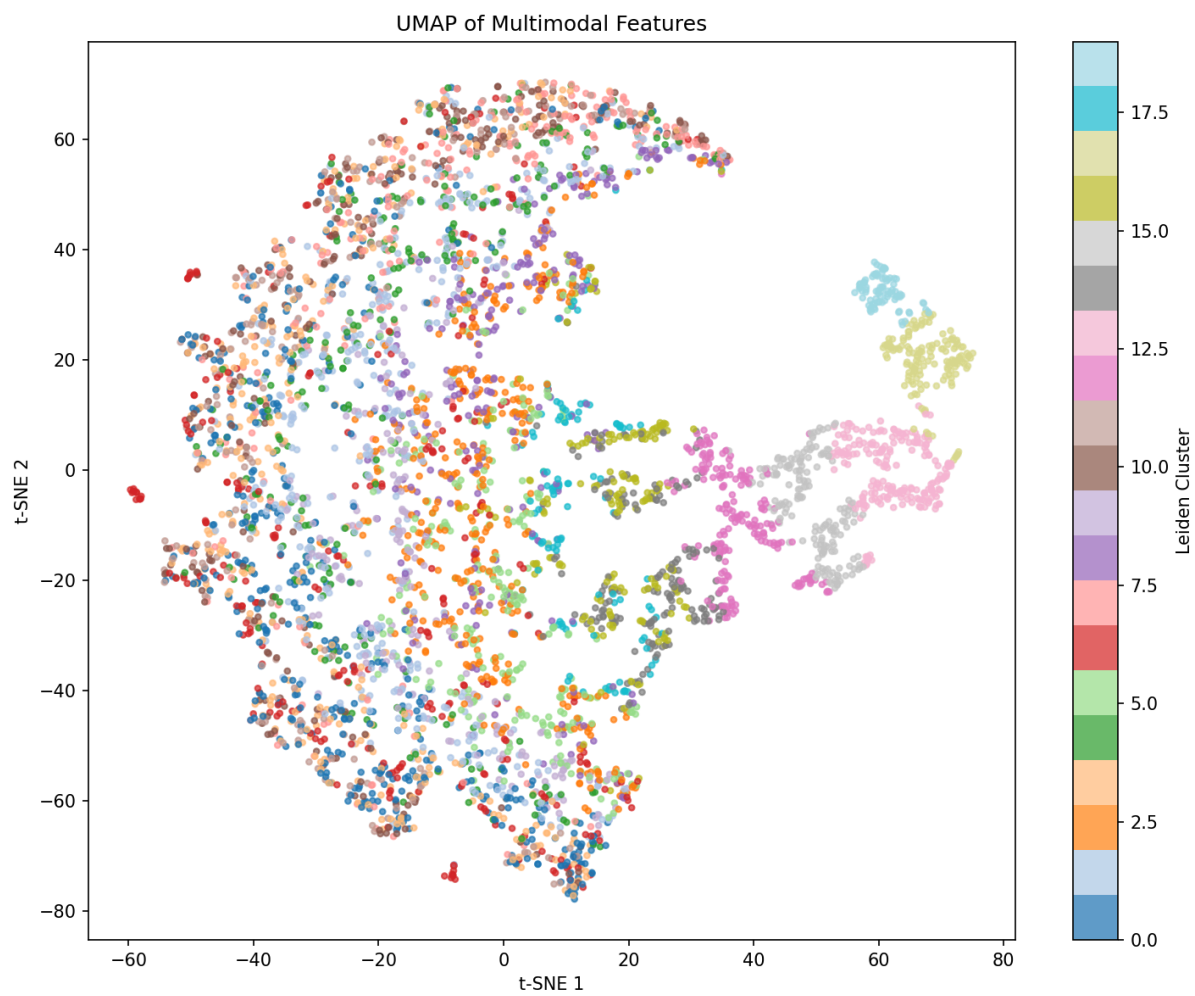


Figure 8: Feature Space UMAP